

YOURSTORY



H2S



Coco CLI

HACKATHON

GCC Edition



Team Name : WU ERA

Team Leader Name : Suja Parvathy

Team Size : 4

Problem Statement : Risk, Fraud and Regulatory Intelligence Copilot

Problem Brief

What real business problem does this solve?

Banking compliance teams cannot efficiently detect, investigate, and report money laundering activity. AML analysts must manually cross-reference transaction data against multiple regulatory frameworks (BSA, FATF, FinCEN, UAE AML/CFT), meet strict filing deadlines (SAR within 30 days, CTR within 15 days), and produce audit-ready documentation — all using siloed tools where transaction monitoring, policy lookup, case management, and regulatory reporting are completely disconnected.

Problem Brief

Who is the target user/persona?

AML Compliance Analysts, BSA Officers, Risk Investigators, and Compliance Managers at banks and NBFCs — professionals who triage alerts, investigate suspicious activity, make filing decisions, and produce regulatory reports daily.

Problem Brief

What is the current pain point and how does this improve it?

Today, an analyst investigating a structuring alert must:

- (1) query the transaction monitoring system,
- (2) manually open a separate policy reference to check the \$10,000 CTR threshold,
- (3) document findings in a case management tool,
- (4) draft a SAR narrative in a word processor, and
- (5) generate a filing in yet another system. RiskLens AI collapses this into a single natural language interface — ask "Which customers show structuring patterns and what are the reporting requirements?" and get a policy-backed answer with transaction evidence, the applicable BSA regulation, a risk assessment, and a recommended action in one response.

Problem Brief

Industry/domain context

Anti-Money Laundering (AML) compliance is a \$50B+ global spend across financial services. Regulatory penalties for non-compliance regularly exceed \$100M. FinCEN received 4.6M+ SARs in 2023 alone. The volume of alerts, complexity of overlapping regulations, and auditability requirements make this one of the most resource-intensive functions in banking.

Architecture Diagram



CoCo CLI skills used and how they connect

Planning: CoCo designed the 13-table data model, AML ontology (6 signal types, risk tiers, case states), and signal-to-report workflow

Development: CoCo authored all SQL DDL, synthetic data generation (500K+ rows), 6 stored procedures (SQL + Python + JavaScript), Semantic View YAML (13 tables, 11 relationships, 7 VQRs), Agent specification, Cortex Search pipeline, and the 83KB Streamlit application

Execution: CoCo deployed tasks, dynamic tables, search indexing, and the Streamlit app to Snowflake

Testing: CoCo debugged the agent via `DATA_AGENT_RUN()`, validated policy search via `SEARCH_PREVIEW()`, iterated the PDF output 3 times, and refined the UI across multiple sessions

Data sources (structured/unstructured)

Structured: 13 relational tables — transactions, customers, accounts, beneficiaries, signals, cases, evidence, countries, risk profiles (500K+ total rows, queryable via Cortex Analyst through Semantic View)

Unstructured: 11 AML policy documents (BSA/AML CTR, SAR Filing, EDD, Wire Monitoring, Sanctions, KYC, UAE AML/CFT, CTF, TBML, Correspondent Banking, PEP) — chunked (1500 chars, 200 overlap) and indexed via Cortex Search

How modular components plug together

The Semantic View is reusable by any Cortex Analyst or Agent consumer. The Cortex Search Service is callable from any application. Stored procedures are independently executable. The Cortex Agent orchestrates both tools through its YAML specification — swap the Semantic View or add policy documents without changing the agent. The Streamlit app calls the agent via `_snowflake.send_snow_api_request` and renders responses generically. The entire system is reproducible from 5 idempotent SQL scripts.

Impact Statement

Measurable outcomes (time saved, accuracy improvement, etc.)

Investigation time: Reduces a typical AML investigation from 4-6 hours (manual cross-referencing across tools) to minutes (single natural language query returns data + policy + recommendation)

Accuracy improvement: Dual-tool enforcement ensures every answer is grounded in both data evidence AND regulatory policy — eliminating the risk of conclusions without policy backing

Signal coverage: 6 deterministic rules running every 5 minutes detect structuring, rapid movement, velocity spikes, unusual geography, dormant activation, and beneficiary bursts — patterns that manual review often misses until batch processing

Report generation: SAR narrative drafting reduced from 1-2 hours of manual writing to seconds via AI generation with policy citations pre-attached

Data quality: Automated 15-minute DQ checks catch integrity issues (NULL rates, orphaned records, referential violations) before they reach analysts

Impact Statement

Scalability potential

- Dynamic tables and stream-triggered tasks scale horizontally with Snowflake compute — the same architecture handles 500K or 50M transactions by adjusting warehouse size
- Semantic View supports adding new tables (e.g., wire transfers, trade finance) without rebuilding the agent
- Cortex Search Service scales by adding new policy documents (GDPR, MiFID II, PSD2) — the agent automatically retrieves relevant regulation
- Multi-jurisdiction: policy documents already cover US (BSA/FinCEN), international (FATF), and UAE — extend to EU, UK, APAC by adding policy documents
- The agent specification supports adding new tools (e.g., sanctions screening, PEP database) alongside existing ones

Impact Statement

How this extends beyond the demo

Production deployment: Connect to real transaction feeds (via Snowpipe or Kafka connector) instead of synthetic data — the detection pipeline, dynamic tables, and agent work identically

Integration: Agent API is callable from any system (case management, SWIFT, core banking) via REST — not locked to the Streamlit UI

Audit and compliance: Every agent response, signal detection run, and DQ check is logged — the pipeline audit log and conversation history provide a complete audit trail for regulatory examination

Model enhancement: Add ML-based anomaly detection as a 7th signal type alongside the 6 deterministic rules — the agent surfaces both equally

Enterprise rollout: Role-based access control, department-specific Semantic Views, and regional policy document sets enable multi-team deployment across a global bank

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THANK YOU

